

1. Write Simple Hello World program using C#.
2. Create a basic four-function calculator. Take two numbers input from command line along with the operation to be performed. Use switch case.
3. Create a basic four-function calculator. Take two numbers input from console along with the operation to be performed. Make a menu driven program, which will continue to execute according to user choice. Use do-while loop.
4. Write a structure Student with following members:
 string name; bool gender; int age;
 int std; char div; double marks;
Provide following functionality:
 Default constructor(?)
 Parameterized constructor
 Get and Set methods
 AcceptDetails() method to accept data from console.
 PrintDetails() method to print data to console.
Check the default access specifier for members of structure.
Keep all data members private and all member functions public.
5. Create a reference of Student's array in Main. Write a function CreateArray() to allocate the array as per user's requirement. Pass it to a function AcceptInfo() to accept student information. Print the array using PrintInfo() function. Also write a function ReverseArray() that reverse the array into a new array. Reprint the modified array in Main.
Note: Do not return array from the function. All four functions mentioned above should be static functions in Main()'s class.
6. Write class Date with following members:
 int day; int month; int year;
Provide following functionality:
 Default constructor
 Parameterized constructor
 Properties: Day, Month and Year
 AcceptDate() method to accept data from console.
 PrintDate() method to print data to console.
 bool IsValid(); method to check validity of date.
 string ToString(); method to return data of object in string format.
Also provide a static member function that returns difference between two date objects in number of years.
Overload "-" operator to perform the same job.
7. Write a class Person with following members:
 string name; bool gender; Date birth;
 string address;

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Name, Gender, Birth, Address and Age(Read Only)

Accept() method to accept data from console.

Print() method to print data to console.

string ToString(); method to return data of object in string format.

Hint: Class will contain Date object for birth and Age will be calculated using static method in Date class.

8. Write a class Employee with following members:

int id; (Auto Generated)

double salary; string designation;

enum DepartmentType dept;

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Id, Salary, Designation, Dept

Accept() method to accept data from console.

Print() method to print data to console.

string ToString(); method to return data of object in string format.

Hint: class must be inherited from Person class. Use static count for auto generating id.

9. Write a class Manager with following members:

double bonus;

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Bonus

Accept() method to accept data from console.

Print() method to print data to console.

string ToString(); method to return data of object in string format.

Hint: class must be inherited from Employee class so that designation is fixed i.e. "Manager".

10. Write a class Supervisor with following members:

int subordinates; (Number of assistants)

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Subordinates

Accept() method to accept data from console.

Print() method to print data to console.

string ToString(); method to return data of object in string format.

Hint: class must be inherited from Employee class so that designation is fixed i.e. “Supervisor”.

11. Write a class WageEmp (Contract basis employee) with following members:

int hours; int rate; (Per hour basis)

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Hours, Rate

Accept() method to accept data from console.

Print() method to print data to console.

string ToString(); method to return data of object in string format.

Hint: class must be inherited from Employee class so that designation is fixed i.e. “Wage”.

12. Write class Company with following members:

string name; string address;

LinkedList empList;

double salaryExpense; (Per month)

Provide following functionality:

Default constructor

Parameterized constructor

Properties: Name, Address, EmpList, SalaryExpense.

Accept () method to accept data from console.

Print () method to print data to console.

CalculateSalaryExpense() method to calculate salary expense per month.

public void AddEmployee(Employee e);

public bool RemoveEmployee(int id);

public LinkedListNode<Employee> FindEmployee(int id);

public override string ToString();

public void PrintEmployees();

13. Write all these classes in a class library “EmplyoeLib”.

14. Create a console application and use the library “EmployeeLib” as private assembly. Console application must be a menu drive program providing facilities of adding employee to company, removing employee, find employee by id, display company info and display all employees.

15. Create a similar console application and use the library “EmployeeLib” as shared assembly.

16. Modify EmployeeLib assembly to provide serialization support. Change the version number of this assembly and re-deploy it into GAC. In menu driven console application add two menus for serialization and deserialization. Use BinaryFormatter class.
17. Modify Company class to use ArrayList of employees rather than LinkedList. Also provide support for SoapFormation. In menu driven console application add two menus for serialization and deserialization. Use SoapFormatter class.
18. Create a static Math class using Sum, Subtract, Multiply, Divide methods. Call these methods from Main() function using uni-cast delegates as well as multicast delegates.
Create another Math class using Sum, Subtract, Multiply, Divide methods as instance methods. Call these methods from Main() function using uni-cast delegates as well as multicast delegates.
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20. Create event "EmpListChanged" in Company class. Event must be fired when employee is added or removed from the company. Event handler in client project should take care of updating salary expense of the company.

21. Create a GUI application using classes in System.Windows.Forms namespace to traverse through employee list of company. Provide buttons First, Prev, Next, Last.
Also provide support for adding, removing and finding employee in the list.
Provide buttons Add, Delete, Remove, Clear.
Provide a menu for Open, Save and Exit. Open and Save menus provide de-serialization and serialization actions respectively.
22. Create a library “ShapeLib” that includes IShape interface, abstract class Shape and Line, Rect, Ellipse classes.
IShape interface provide abstract method for drawing a shape.
void Draw(Graphics gr);
Create a new class Shape inheriting interface IShape. Provide data members:
 Point ptStart;
 Point ptEnd;
 Color penColor;
 int penWidth;
 DashStyle penStyle;
Also write constructors and corresponding properties, but Draw() method kept abstract in this class.
Line, Rect and Ellipse class is inheried from Shape class and implement Draw() method.
- Create a GUI application, which have following menus.
Shape → Line, Rectangle, Ellipse
Color → Red, Green, Blue, Custom
GDI → Pen.
Implement all the above functionalities.
When Pen menu is clicked, a dialog box must be opened that takes PenWidth and PenStyle from the user. The next shape drawn should get these attributes.
23. Modify ShapeLib to provide generate “ColorChanged” event. The event should be fired, when color of any shape (already drawn) is modified by the user.
In GUI application, Add a ListView that keeps description of every drawn shape. Whenever any shape is drawn its StartPoint, EndPoint, Color, Width is stored in list view. If user clicks on any shape in list view, a color dialog should appear, which allows user to change color of that shape. At this time ColorChanged event must be fired, that will cause the shape to be redrawn with new color.
24. Modify above application as a MDI window. Each MDI child shows different drawing. Manage multiple child windows using File → New, Close, Exit. Menu.
25. In above application add File → Save, Open. Options to save the shapes into an xml file using SoapFormatter.

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27. Write your own Stack class using generics. Implement the interface IEnumerable.
28. Write your own LinkedList class. Implement Indexer and Iterator for this class.
29. Add custom attribute “Readme” to EmployeeLib.Company class. This attribute provide information about author of this class and date of creation of this class.
30. Write a GUI application to retrieve the information of any assembly using Reflection technique. Display all this information in tree view like “ILDAsm.exe” utility of .net framework.
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